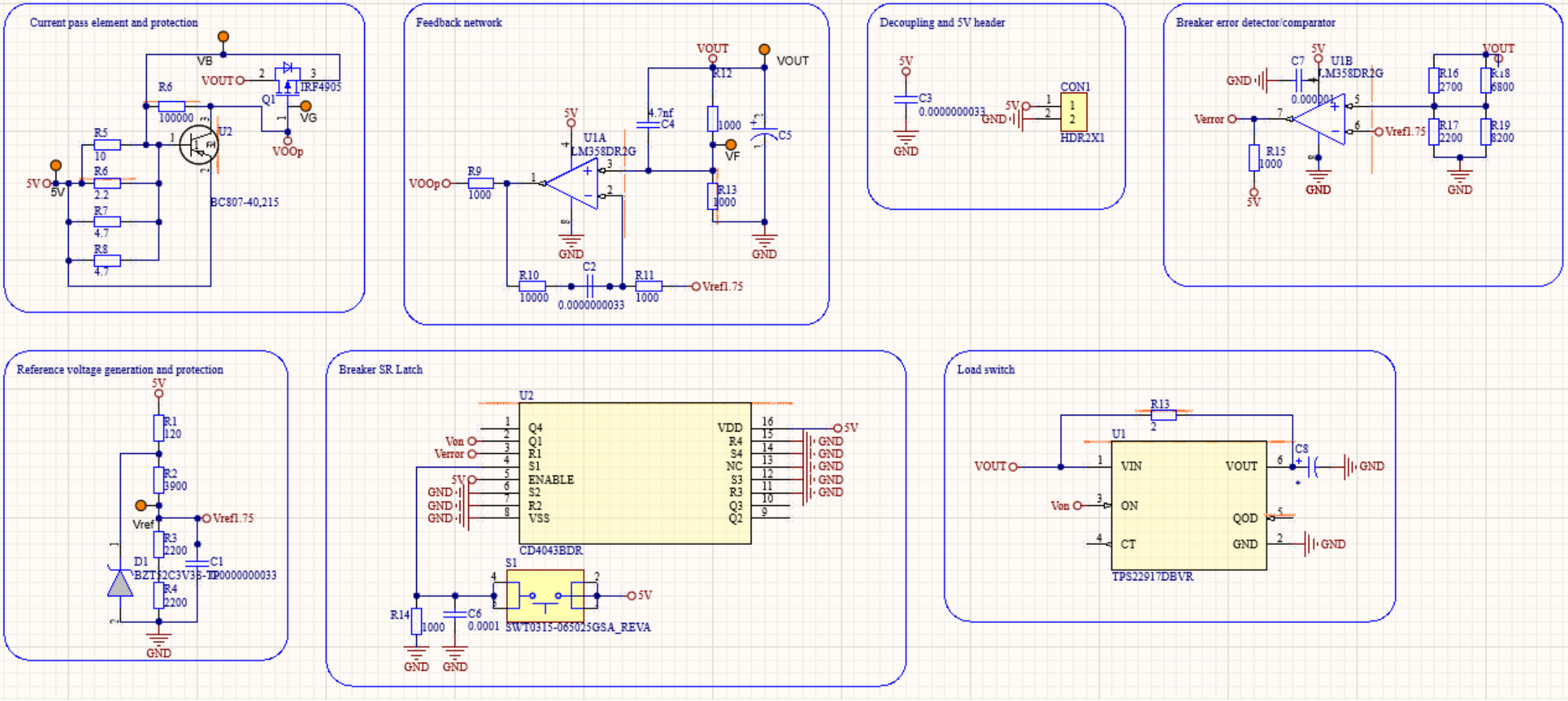
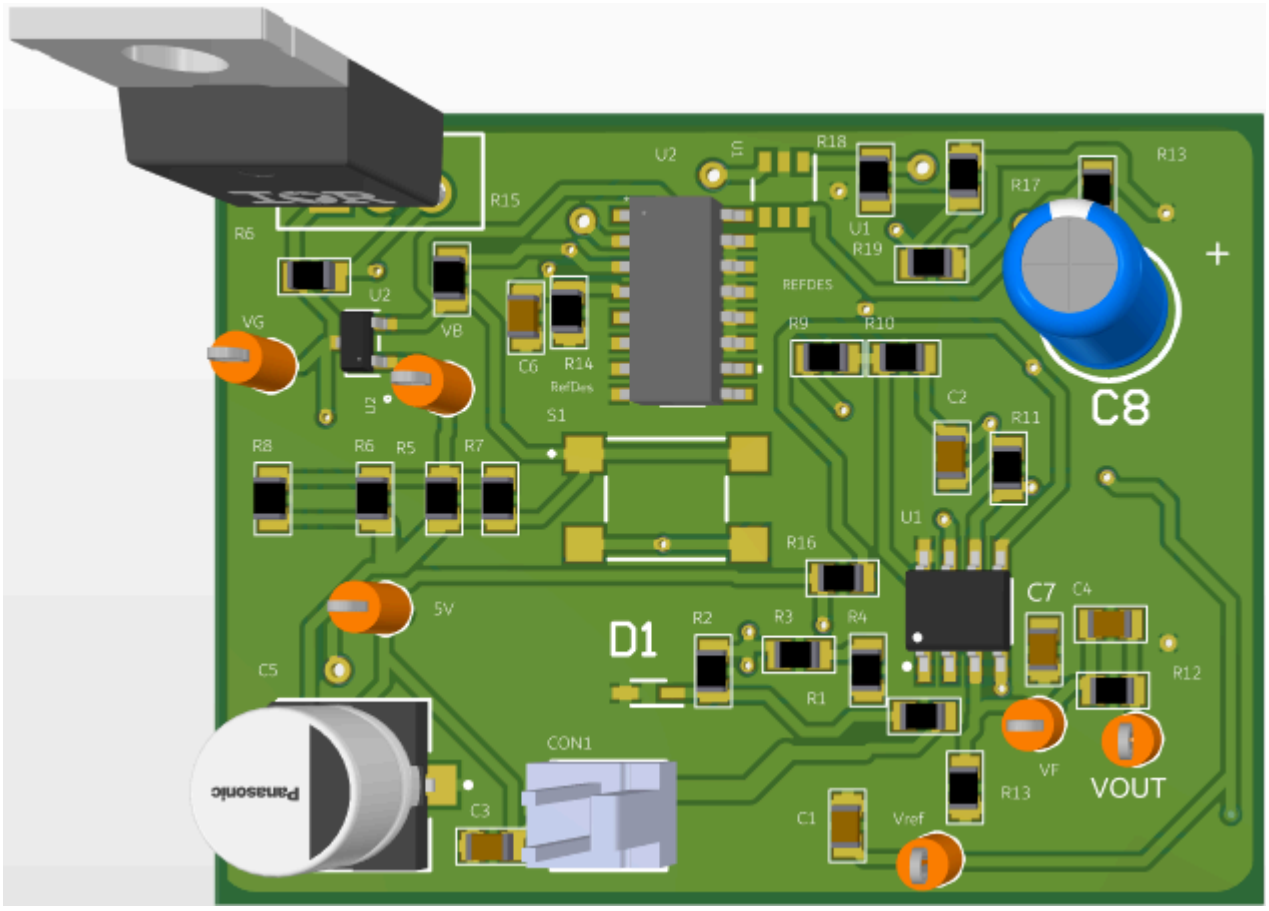
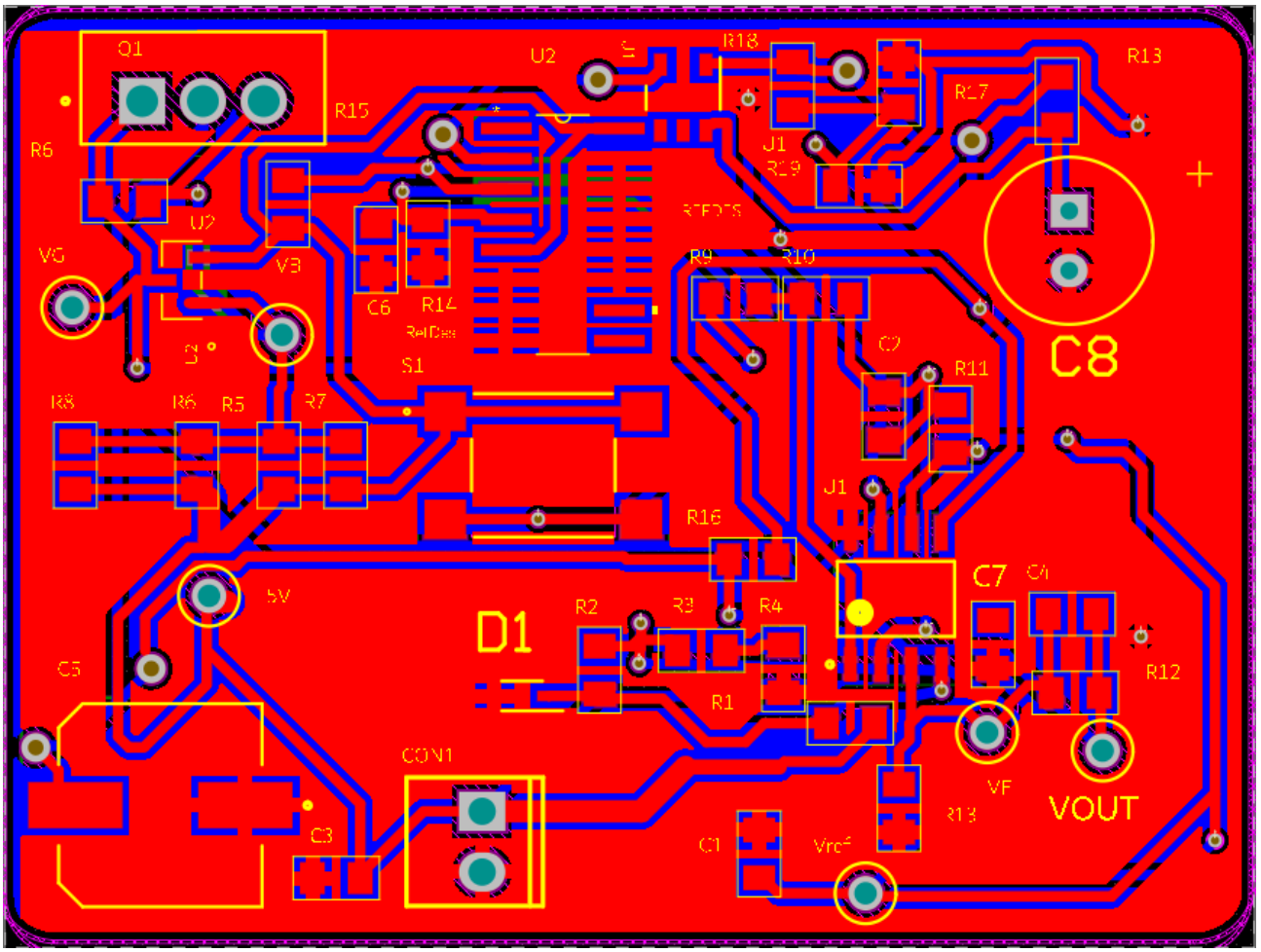
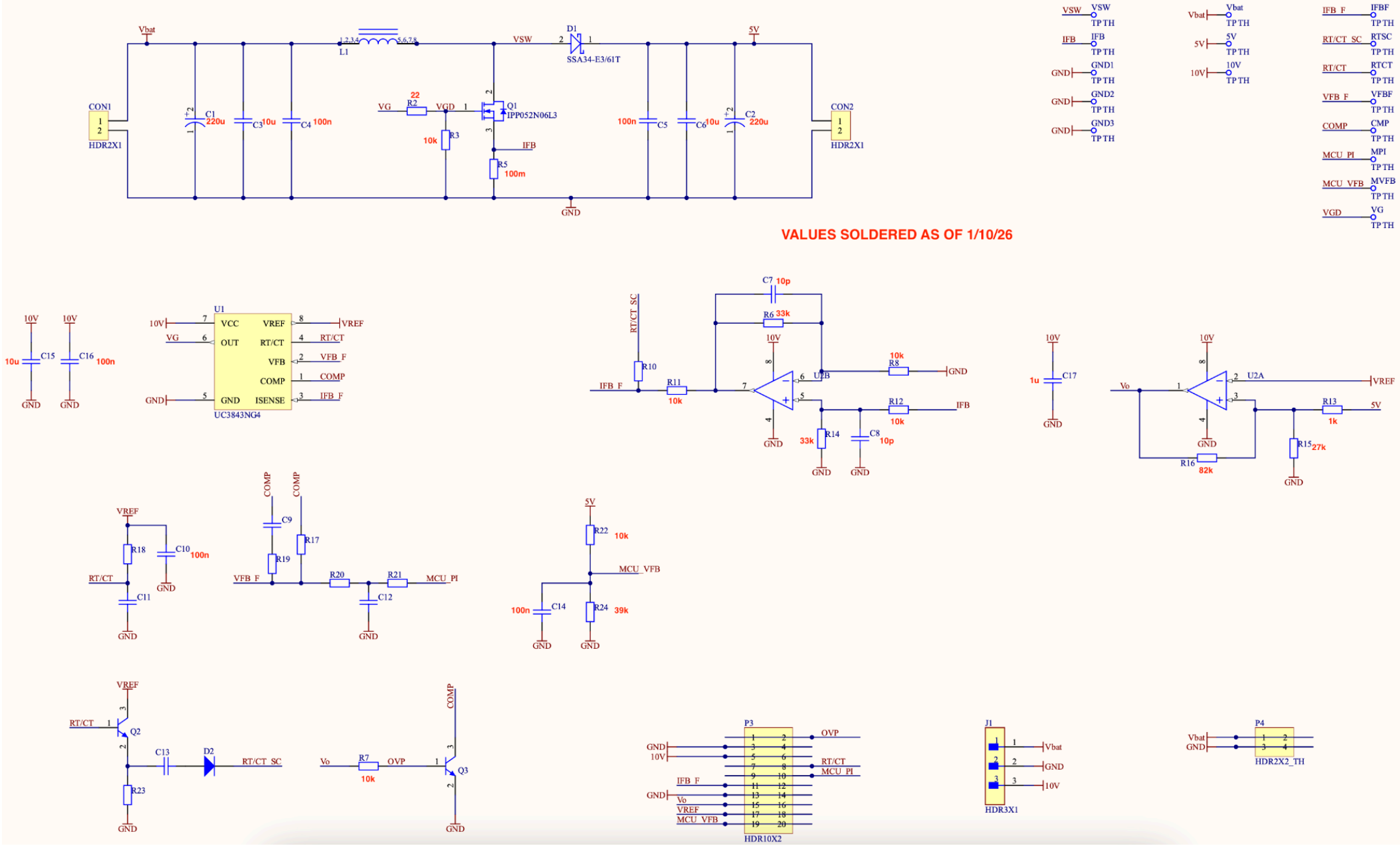
Altium Schematic

ELECTENG 311 Team 9 Linear Charger and Breaker Schematic




Q1
R15
R6
U2
VG
VB
C6
R14
RefDes
S1
R18
U1
R19
REFDES
R9
R10
R17
R13
C8
C2
R11
R8
R5
R7
U1
R16
C7
C4
5V
D1
R2
R3
R4
R1
R12
C5
CON1
C3
C1
Vref
VF
VOUT
Panasonic


Q1
R15
R6
U2
VG
VB
C6
R14
RefDes
S1
R18
J1
R19
REFDES
R9
R10
R17
R13
C8
C2
R11
R8
R5
R7
J1
R16
C7
C4
5V
D1
R2
R3
R4
R1
R12
C5
CON1
C3
C1
Vref
VF
VOUT


Vbat
L1
VSW
D1
SSA34-E3/61T
5V
CON1
HDR2X1
C1 220u
C3 10u
C4 100n
VG
R2 22
VGD
Q1
IPP052N06L3
R3 10k
IFB
R5 100m
GND
100n C5
C6 10u
C2 220u
CON2
HDR2X1
VSW
IFB
GND1
GND2
GND3
Vbat
5V
10V
IFB_F
RT/CT_SC
RT/CT
VFB_F
COMP
MCU_PI
MCU_VFB
VGD
VALUES SOLDERED AS OF 1/10/26
10V
C15 10u
C16 100n
U1
VCC
OUT
GND
VREF
RT/CT
VFB
COMP
ISENSE
UC3843NG4
C7 10p
R6 33k
R10
R11 10k
U2B
R8 10k
R12 10k
R14 33k
C8 10p
C17 1u
U2A
R13 1k
R15 27k
R16 82k
VREF
R18
C10 100n
C11
C9
R17
R19
VFB_F
R20
R21
MCU_PI
C12
R22 10k
C14 100n
R24 39k
Q2
C13
D2
R23
R7 10k
OVP
Q3
P3
HDR10X2
J1
HDR3X1
P4
HDR2X2_TH

Boost Converter

Boost Converter Testing

1. Over-Voltage Protection (OVP)

1.1 Test plan

Aim: verify the OVP trip level, the release level (hysteresis) and the response speed of the comparator acting on the COMP node.

- UC3843 fitted, 12 V rail powered, so the protection circuit and COMP node are live.
- A bench DC supply drives the 5 V test point in place of Vout. It was swept slowly through the threshold in both directions.
- Oscilloscope (R&S RTB2002): C1 probes the Vout test point, C2 probes COMP. Single-shot captures were triggered on the COMP edge, with cursors on the C1 level at the start of the COMP transition.
- Trip: C1 raised slowly through about 5.25 V. Release: C1 lowered slowly through about 5.1 V. Each direction was captured several times to check repeatability.
- Speed (1): measured as the time between the COMP (C2) pin beginning its drop to the time it crosses the zero axis)
- Speed(2): Use a pulsed input to measure the time taken for the whole system to react from OVP crossing to the time the COMP pin drops to zero.

1.2 Results

When C1 rises past the upper threshold, COMP is pulled from about 0.8 V down to ground (the trace settles slightly below 0 V in the short term, later settling at zero). It stays low until C1 falls back below a lower threshold, then returns to about 0.7–0.9 V. This is the expected hysteretic behaviour.

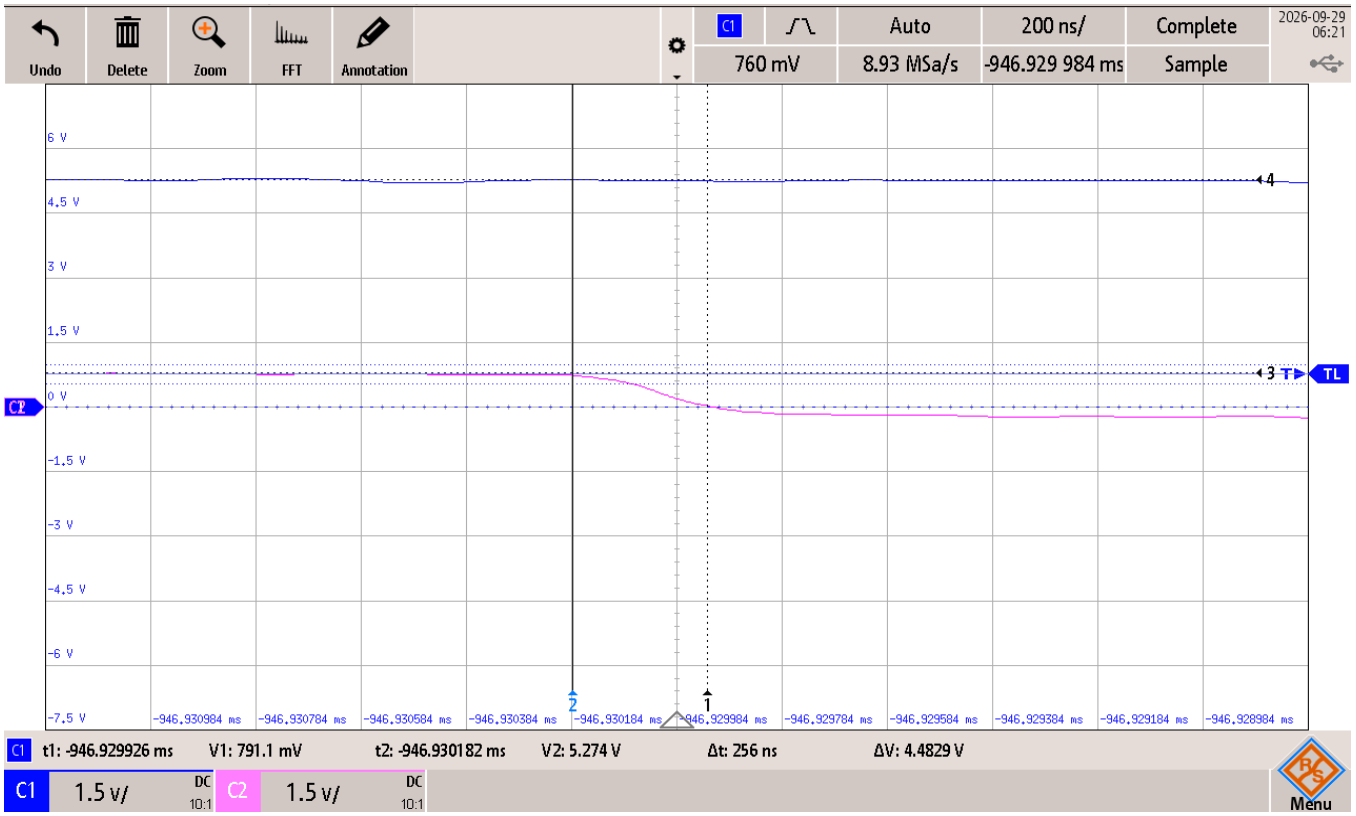


Figure 1: OVP trip. C2 (COMP) falls as C1 reaches 5.274 V. Cursor Δt = 256 ns is the time taken for the COMP pin to reach zero.

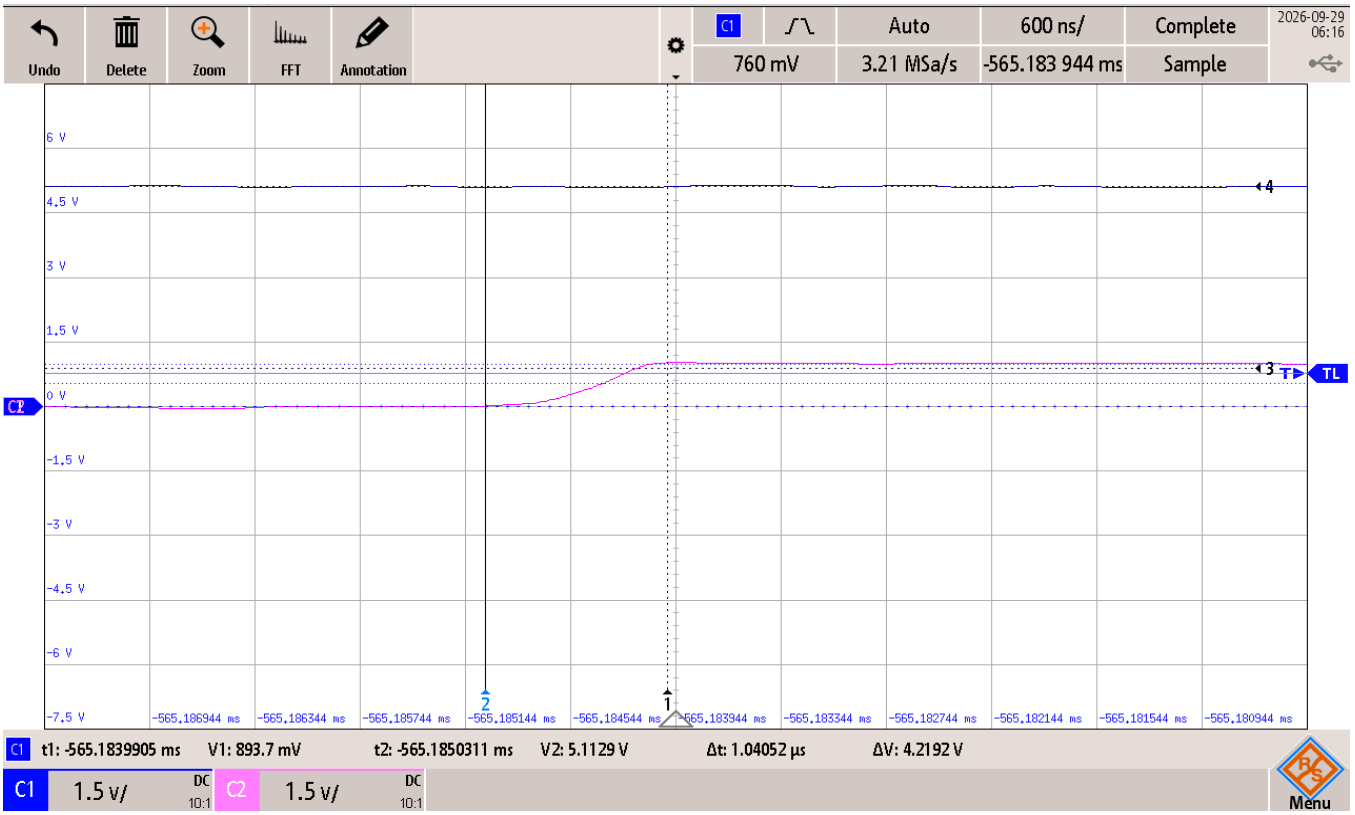


Figure 2: OVP release. C2 (COMP) rises as C1 falls to 5.113 V. Cursor Δt = 1.04 µs is the time taken for the COMP pin to climb back to its stable operational voltage.

Quantity	Oscilloscope	Theory	Deviation from Theory
Trip level (C1 rising)	5.274 V	5.246 V	+0.54%
Release level (C1 falling)	5.113 V	5.1 V	+0.25%
Hysteresis	0.161 V	0.146 V	+10.2%
COMP pin, fall time	256 ns		
COMP pin, rise time	1.04 µs		

Both thresholds are repeatable to about ±10 mV.

2. Current Measurement (OpAmp and Filter)

2.1 Test plan

Aim: characterise the DC gain and the dynamic response of the OpAmp stage and its output filter.

- UC3843 is removed and 12 V rail supplied for the OpAmp only, so the feedback control does not interfere with the clean oscilloscope generated signal that is driven directly to Vg.
- An arbitrary load and signal are applied to the boost converter, in this case,
    - Vbat = 3.5 V, load 16.25 Ω.
    - Function generator: 5 Vpp, 2.5 V offset, 45 % duty, fs = 100 kHz.
- Probes data from IFB (sense signal before the amplifier) and IFBF (amplified and filtered output) are then plotted to demonstrate the amplification and response.

2.2 Results

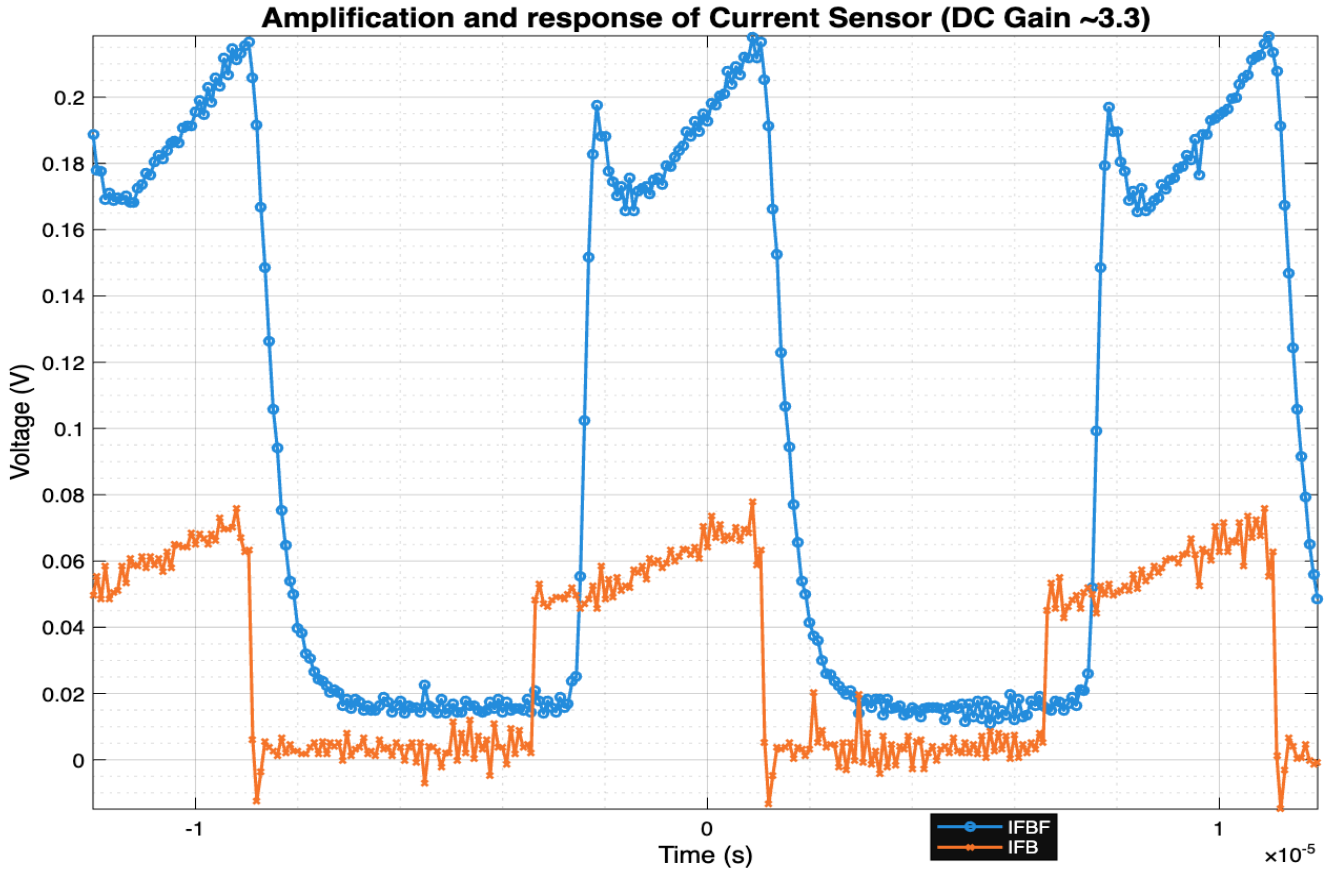


Figure 3: IFB and IFBF over two 10 µs periods. IFB is the raw sense signal and IFBF is the amplified, filtered output.

2.3 Discussion

Gain. The gain is approximately 3. At the end of the on-phase, IFB is 69.7 mV and IFBF is 207.4 mV, a ratio of 2.98. The on-phase averages agree (IFB 62 mV, IFBF 185 mV, ratio 2.97). The DMM reading is 23.5 mV in and 77 mV out, a gain of 3.28, in agreement with the design value of about 3.3. Removing the average floor offsets (IFBF 14.5 mV, IFB 3.4 mV) gives about 2.95 dynamically, so the DC gain is about 3.3 and the gain seen at 100 kHz is about 3.

Delay and response of filter. The filter capacitor, together with its associated resistor, forms a low-pass. It removes the switching noise visible on IFB (the raw signal is noisy, IFBF is smooth), and it is also what causes the delay. IFBF follows the IFB rising edge about 1.07 µs later, with a 0.28 µs rise time, and follows the falling edge about 0.24 µs later, with a 0.88 µs fall time. A larger capacitor gives cleaner IFBF but a longer delay, which delays the current-limit action. The 1 µs delay is small against the 10 µs period, giving an acceptable trade-off.

3. Open Loop Various Load Testing

3.1 Test plan

Aim: measure the open-loop duty cycle needed to hold a 5 V output, and the resulting efficiency, across a range of loads at both ends of the battery range, with no form of feedback control.

- Vin = 2.5 V and 3.5 V; load resistance 5, 16.25, 27.5, 38.75 and 50 Ω.
- UC3843 fitted and powered from 12V rail while VFB and ISENSE are pinned to ground so as to not interfere with the signal generated duty cycle being driven directly into RT/CT
- At each point the duty cycle was adjusted until Vout (probed with an oscilloscope on the 5V (Vout) testpoint)) was 5 V. Pin and Pout were recorded from the DC power supply and electronic load to determine efficiency = Pout / Pin.
- The duty cycle is set on the signal generator, which is inverted relative to the real duty cycle, so the real duty is 100% minus the generator setting. The real duty is plotted.

3.2 Results

Vin (V)	Vout (V)	Load (Ω)	SG duty (%)	Real duty (%)	Pin (W)	Pout (W)	Eff. (%)
2.5	4.7*	5	25	75	6.95	3.3	47
2.5	5	16.25	41	59	1.71	1.35	79
2.5	5	27.5	43	57	1.00	0.83	83
2.5	5	38.75	44	56	0.70	0.60	86
2.5	5	50	44	56	0.55	0.48**	87
3.5	5	5	55	45	5.16	3.4	66
3.5	5	16.25	61	39	1.61	1.32	82
3.5	5	27.5	63	37	0.97	0.84	87
3.5	5	38.75	64	36	0.71	0.62	87
3.5	5	50	65	35	0.55	0.49**	89

* At 2.5 V and 5 Ω the converter failed to reach 5 V; this point was taken at approximately 4.75 V.
** At the lightest load (50 Ω) the circuit delivers 0.48 W at 2.5 V and 0.49 W at 3.5 V, which is 4 % and 2 % below the 0.5 W minimum.

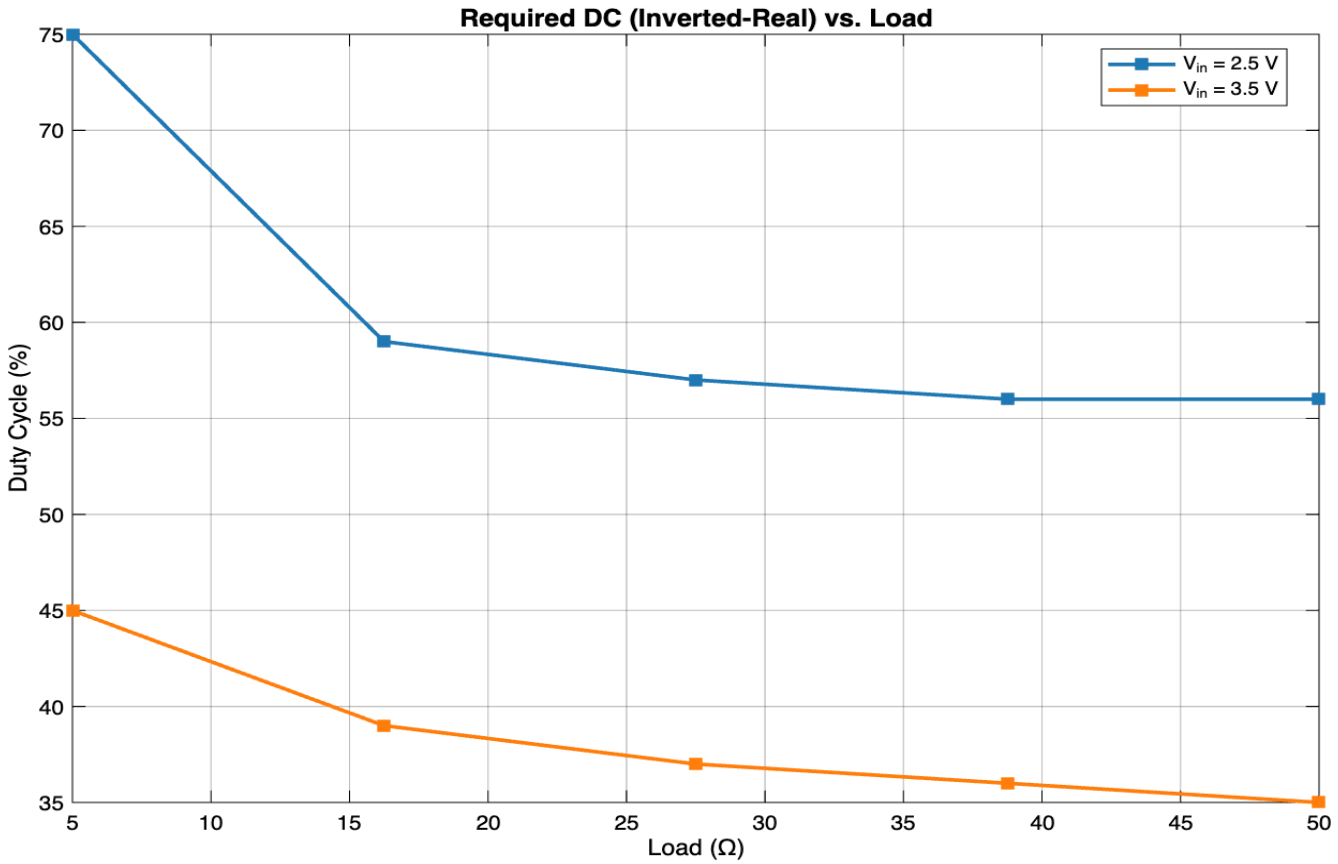


Figure 4: Required (real) duty cycle versus Load.

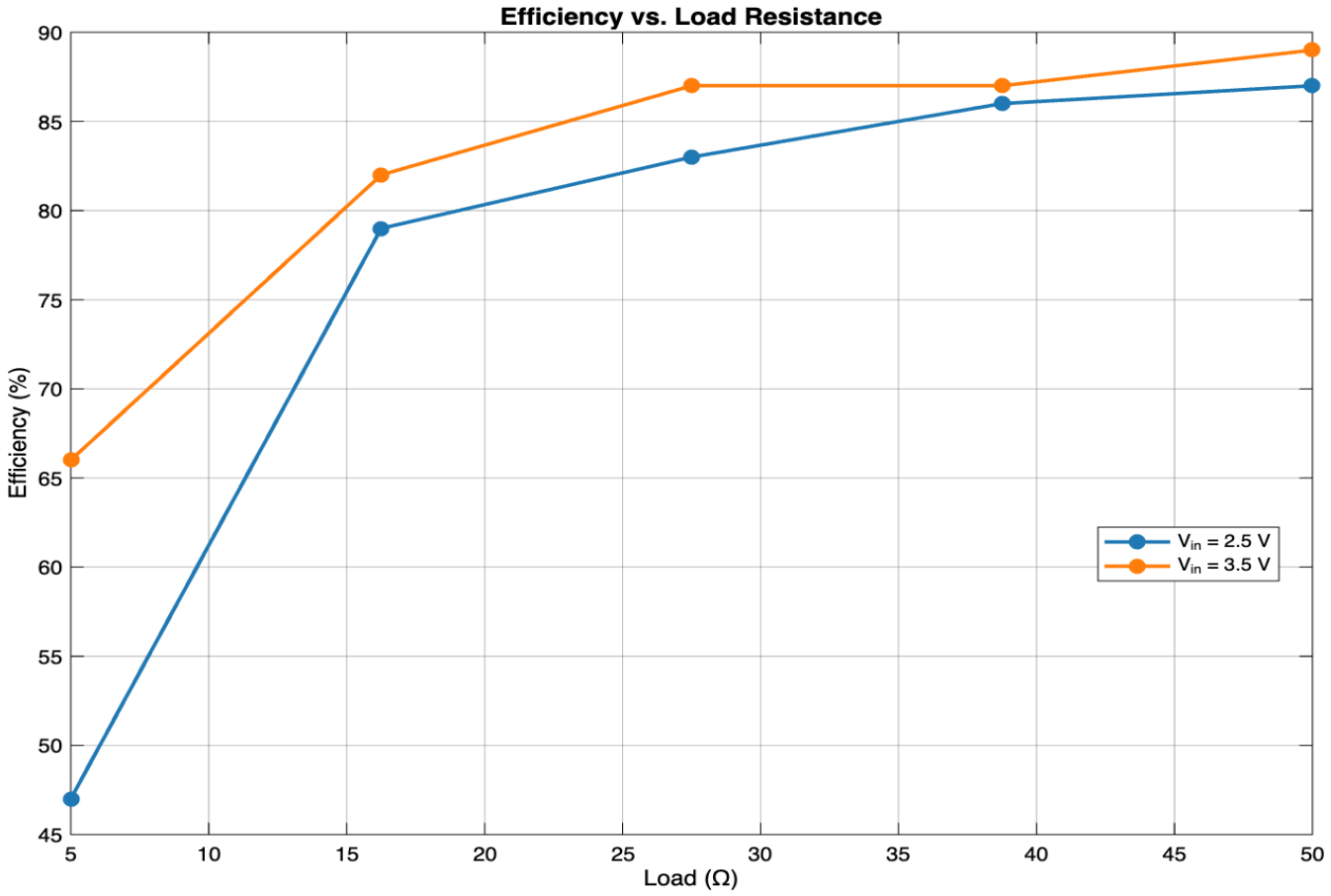


Figure 5: Efficiency versus load resistance.

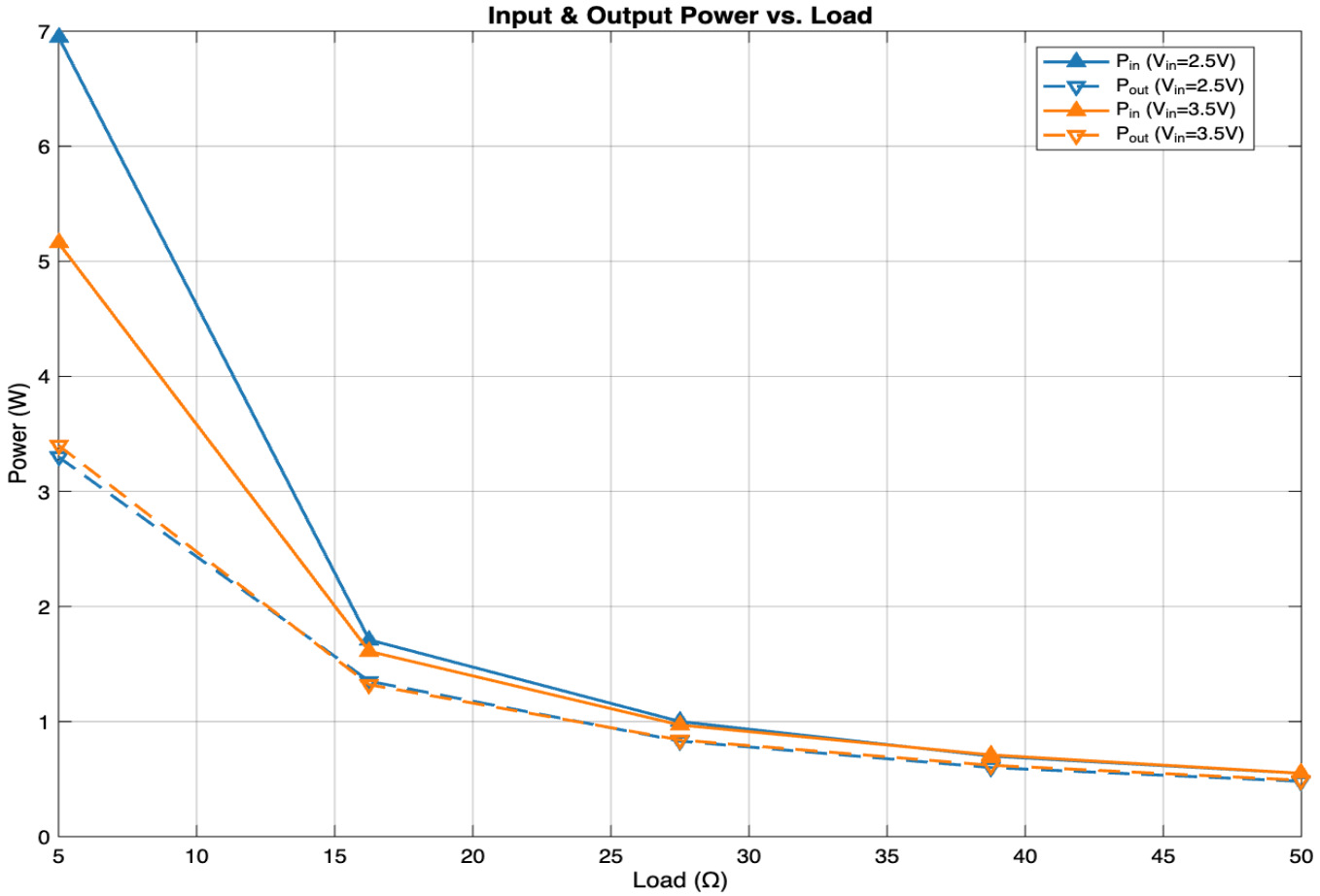


Figure 6: Input and output power versus load.

3.3 Discussion

The required duty cycle falls as the load resistance rises, because less current is drawn and less is lost in the converter. It is also much higher at 2.5 V than at 3.5 V. For an ideal boost converter a 5 V output needs 50 % duty at 2.5 V and 30 % at 3.5 V. The measured values at light load (56 % and 35 %) are about 5–6 points above this, which is consistent with losses that the duty cycle has to make up.

Efficiency rises with load resistance, from 47 % (2.5 V) and 66 % (3.5 V) at 5 Ω to 87 % and 89 % at 50 Ω. At 5 Ω the input power is highest (6.95 W at 2.5 V, an input current of about 2.8 A, against 5.16 W and about 1.5 A at 3.5 V), so resistive losses dominate. The higher input voltage is more efficient at every load, by 19 points at 5 Ω but only 1–4 points from 16.25 Ω upwards. The 5 Ω, 2.5 V case is the worst: even at 75 % duty the output only reached about 4.75 V, at this operating point the losses grow with the input current, so raising the duty further gives diminishing returns.

Over the 16.25–50 Ω range the converter holds 5 V at 79–89 % efficiency. The shortfall in power at 50 Ω (footnote **) is small, at 2–4 % below 0.5 W.

Charger

Breadboard vs Plecs simulated results:

After finalising our component values we decided to breadboard our charger to get a rough idea of whether our charger design was feasible. After constructing our design on the breadboard We found the results at the nodes shown in the graph matched up reasonably well with our expected simulation results. We did however see the output voltage measure up around 200mV lower than the expected value which we suspect to be a breadboard/jumper wire issue rather than a fault in our design.

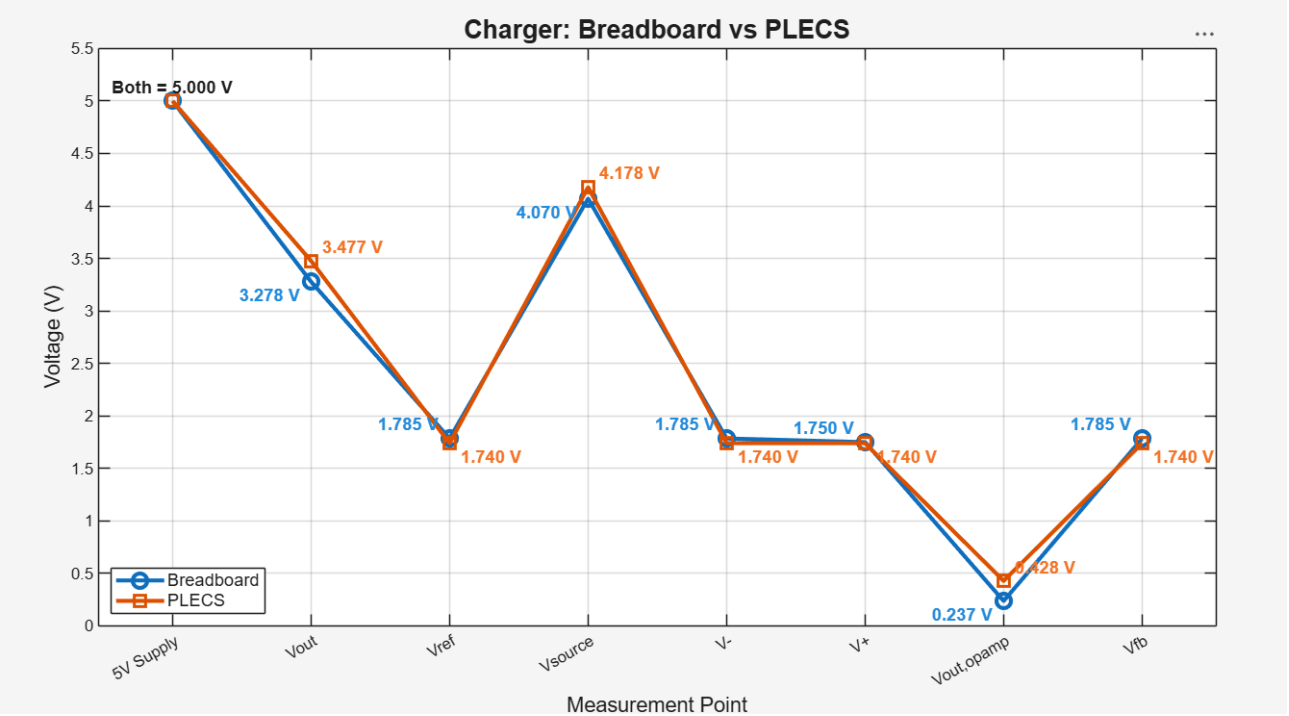


Figure 7: Breadboard vs Simulation results at multiple testing points

Test Plan for PCB:

For our pcb our test plan involves us testing the voltage at each node and comparing it to our simulation results. The most important test points are the ones shown in the graph above which we aim to replicate results as close to the simulation as possible.

Apart from the generic functionality of the charger we aim to also test various safety features in our charger. The vref protection using a zener diode serves the purpose of protecting the two opamps in our circuit in case there is a sudden increase in the input voltage. The voltage ripple checks will be done to ensure a voltage ripple above our comparator threshold for our breaker does not false trigger due to ripples in the output voltage.

Linear Regulator:

Test	Test Method	Expected Results
Test Points	Measure the 5 V supply, Vout, Vref, Vsource, V−, V+, op amp output and Vfb on the assembled PCB. Compare these values against the breadboard measurements and PLECS simulation.	PCB measurements should be close to the breadboard and simulated values, with Vout regulating at approximately 3.5 V.
MOSFET Temperature	Operate the charger under load and measure the Mosfet temperature after it has stabilised. Repeat at different load conditions.	MOSFET temperature should remain within its safe operating temperature range.

Vout Voltage Ripple	Measure Vout using the oscilloscope and record the peak to peak voltage ripple.	Vout should remain stable around 3.5 V with less than 5% deviation.
Vref Protection	Increase the supply to the Vref circuit while monitoring Vref to test the Zener protection.	The Zener should clamp Vref and prevent it from increasing significantly above its intended value.

Current limiting / Vout expectations:

After refining our pcb I have found our current limit is set perfectly at 0.5A. I have noticed that reducing the load during our current limiting resistance range does cause the current to drop or vary rather than stay at the hard 0.5A limit. To test this I will measure the voltage at all 3 terminals of the mosfet as well as the bjt to find out if there is any sudden increase in a particular node before determining whether this problem is an issue which is fixable or not.

Test	Test Method	Expected Results
Voltage Regulation	Begin with a high load resistance and progressively decrease the load resistance. Measure Vout and Iout at each load and compare the results against the PLECS simulation.	Vout is expected to remain approximately 3.5 V down to a 7 Ω load, corresponding to 0.5 A.
Current Limit	Continue decreasing load resistance until the current limit activates .	Output current should be limited to approximately 0.5 A rather than continuing to increase.
Current Limited Output	Test several load resistances below 7 Ω and record Vout and Iout.	Current should remain approximately 0.5 A, while Vout decreases as load resistance is reduced.

Over Voltage Breaker:

Test	Test Method	Expected Results
Overvoltage Comparator	Slowly increase the voltage applied to the Vout sensing input while monitoring the comparator output. Record the voltage at which the comparator changes state.	The comparator output should switch towards the 5 V rail when Vout reaches approximately 3.7 V.
Sr Latch Reset	Increase the sensed Vout above the overvoltage threshold to trigger the comparator and reset the SR latch. Monitor the latch output and breaker state.	The latch should enter its reset state and turn the breaker off, isolating the supercapacitor from the rest of the circuit.
Sr Latch Set	With the latch in its reset state and the breaker off, press the push button connected to the Set input.	The SR latch should return to Q = 1, turning the breaker back on and reconnecting the circuit.
Breaker Functionality	Apply voltages below and	For input voltages of 3.7 V

	above the 3.7 V overvoltage threshold and measure the voltage before and after the breaker.	and below, the breaker should remain connected and the output should approximately equal the input voltage, allowing for a small voltage drop or ripple. Above 3.7 V, the breaker should disconnect the supercapacitor from the rest of the circuit, giving an output of approximately 0 V.

Task Allocation:

Kyle - Charger schematic / breadboarding / testing
Bruce - Charger pcb / breadboarding / testing
Alex - Boost Converter construction / testing